

PATIENT CASE QUESTIONS

1. Non–Small Cell Lung Cancer

Patient presentation

Chief Complaint

–I have been coughing up blood.‡

HPI

This 66-year-old woman presents to her PCP with complaints of a dry, nonproductive cough for 2.5 months, dyspnea on exertion, and hemoptysis for 1 week.

PMH

Dyslipidemia

HTN

Anemia of unknown etiology × 1 year

Type 2 DM

Purified protein derivative (PPD) (–)

FH

Father died of colorectal cancer at age 68

Aunt died of breast cancer at age 70

SH

Married, lives with son and daughter; 30 pack-year cigarette smoking history (approximately 1 ppd × 30 years); occasional ETOH use; no known recent exposure to TB

Meds

Folic acid 1 mg PO daily

Ferrous sulfate 325 mg PO TID

Simvastatin 20 mg PO daily

Metformin 500 mg PO BID

Pantoprazole 40 mg PO daily

Allergies (All)

Penicillin (rash)

Sulfa (rash)

ROS

(+) For pulmonary symptoms as noted in HPI; no headaches, dizziness, or blurred vision

Physical Examination

Gen

Mildly overweight Caucasian woman in slight distress. ECOG performance status of 1.

VS

BP 169/100 mm Hg, P 90 bpm, RR 30, T 37.2°C; Wt 82 kg, Ht 5'6" Skin

Patches of dry skin; no lesions

HEENT

PERRLA; EOMI; fundi benign; TMs intact

Neck/Lymph Nodes

No lymphadenopathy; neck supple

Lungs

Wheezing in RUL; remainder of lung fields clear

Heart RRR; normal S1, S2, slight systolic murmur on le  lateral side

Abd

Soft, nontender; no splenomegaly or hepatomegaly

Genit/Rect

Normal female genitalia; guaiac (–) stool

Neuro

A&O × 3; sensory and motor intact, 5/5 upper, 4/5 lower; CN II–XII intact; (–) Babinski

Labs RRR; normal S1, S2, slight systolic murmur on le  lateral side

Abd

Soft, nontender; no splenomegaly or hepatomegaly

Genit/Rect

Normal female genitalia; guaiac (–) stool

Neuro

A&O × 3; sensory and motor intact, 5/5 upper, 4/5 lower; CN II–XII intact; (–) Babinski

Labs

Na 138 mEq/L	Hgb 10.7 g/dL	Ca 9.7 mg/dL
K 3.6 mEq/L	K 3.6 mEq/L	Mg 2.0 mg/dL
Cl 101 mEq/L	Plt $255 \times 10^3/\text{mm}^3$	
CO ₂ 23 mEq/L	WBC $9.9 \times 10^3/\text{mm}^3$	
BUN 11 mg/dL		
SCr 1.1 mg/dL		
Glu 182 mg/dL		

Chest X-Ray

PA and lateral views reveal a possible mass in right upper lobe

FIGURE 1.

Chest x-ray with PA (A) and lateral (B) views showing a possible mass in the right upper lobe (arrows).



Assessment

A 66-year-old woman with new-onset hemoptysis is admitted for workup of a possible lung mass.

She has anemia and a history of dyslipidemia, DM type 2, and HTN.

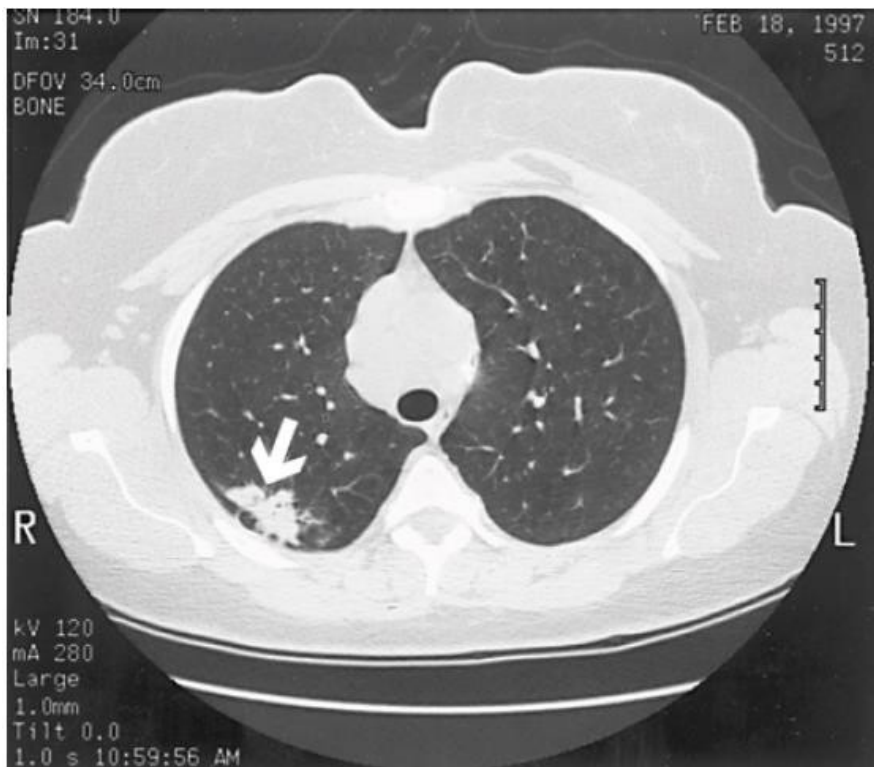
Clinical Course (PART 1)

The patient was further evaluated for lung cancer on an outpatient basis. A bronchoscopy (with biopsy) was performed that identified squamous cell carcinoma. The chest CT scan revealed a 3- × 2-cm right lung mass

(Fig.2). A mediastinoscopy was performed to determine the resectability of the tumor. The mediastinoscopy and biopsy revealed unresectable stage IIIB (T2N3) NSCLC with metastases to the contralateral mediastinal nodes. A brain MRI was negative for disease. Tumor EGFR status was positive by IHC

FIGURE 2.

CT scan of the chest revealing a 2.5- × 2-cm right lung mass (arrow).



QUESTIONS

1. What subjective and objective information indicates the presence of non–small cell lung cancer (NSCLC) in this patient and assess the severity of the NSCLC based on the subjective and objective information available?
2. Create a list of the patient’s drug therapy problems and prioritize them. Include assessment of medication appropriateness, effectiveness, safety, and patient adherence.
3. What feasible pharmacotherapeutic alternatives are available for treating stage IIIB NSCLC?
4. Create an individualized, patient-centered, team-based care plan to optimize medication therapy for this patient’s drug therapy problems. For NSCLC chemotherapy, include specific drugs, dosage forms, doses, schedules, and durations of therapy.
5. In addition to chemotherapy, what measures should be taken to ensure the tolerability of the NSCLC regimen and to prevent adverse effects?
6. Calculate the patient’s BSA, creatinine clearance, and the amount of each drug to be administered based on the regimen chosen.
7. Would the treatment plan for this patient change if she had presented initially with stage IIIB NSCLC that was positive for a sensitizing EGFR mutation? If so, what would the treatment be?

Follow-up: Monitor and Evaluate

8. What clinical and laboratory parameters should be used to evaluate the therapy for achievement of the desired therapeutic outcome and to detect or prevent adverse effects?

CLINICAL COURSE (PART 2)

The patient’s subsequent courses were complicated by the occurrence of DVT, weight loss, neutropenic fever, anemia, nausea/vomiting, and infections. At one point, she presented with a serum calcium level of 11.3 mg/dL and an albumin of 1.2 g/dL, with symptoms of weakness, confusion, nausea, and vomiting.

FOLLOW-UP QUESTIONS

9. Calculate the patient’s corrected calcium level and provide an interpretation of that value and what treatment modalities may be used to correct hypercalcemia?

CLINICAL COURSE (PART 3)

Seven weeks after beginning the new chemotherapy regimen, the patient presents to the ED with complaints of headache and mental status changes as per the patient's husband and caregiver. An MRI of the head reveals multiple lesions, most likely brain metastases.

FOLLOW-UP QUESTIONS

- 10.** Briefly discuss options (drug and nondrug) to treat brain metastases.
- 11.** What is the role of anticonvulsant agents in the setting of brain metastases?

2. Non-Hodgkin Lymphoma

Patient presentation

Chief Complaint

–What’s the next step for my lymphoma?||

HPI

Alemu is a 58-year-old man who presents to his oncologist’s office for recommendations about treatment of a newly diagnosed diffuse, large B-cell lymphoma. He had been in relatively good health other than his longstanding hypertension, hypercholesterolemia, and GERD. He initially presented to the ED 1 week ago with new-onset left lower extremity edema. He was then hospitalized for further evaluation and treatment. At that time, he stated that he had lost weight over the past few months. Vital signs on admission included elevated temperature of 100.8°F (38.2°C). Physical examination findings were significant for a round, protuberant abdomen that was soft on the right side but firm on left and 3+ pitting edema of the left lower extremity from foot to knee. Laboratory findings were significant for serum creatinine 1.9 mg/dL, serum calcium 11.6 mg/dL, and hemoglobin 10.6 g/dL. A renal ultrasound revealed severe hydronephrosis on the left side and displacement of the bladder with compression, possibly from a mass. A CT scan without contrast of the chest, abdomen, and pelvis was performed. Results showed multiple enlarged lymph nodes, including several mediastinal nodes, paraesophageal nodes, and retrocrural nodes. A large, confluent soft tissue mass was found, extending through the retroperitoneum and involving the left pelvis. The mass encased the distal left ureter. The mass in its widest dimension was 16.3 cm. A diagnosis of lymphoma was presumed based on the extent of lymph involvement. Urologists placed a left ureteral stent and performed ultrasound-guided biopsy of the mass. Pathology showed diffuse large non-Hodgkin B-cell lymphoma. Upon confirmation of diagnosis, a bone marrow biopsy was performed. Results were not available during hospitalization. The patient was insistent on discharge to attend his daughter’s wedding, so treatment was deferred after initial consultation with an oncologist in the hospital. His creatinine and calcium values were normalizing at the time of discharge.

PMH

HTN × 10 years

Hypercholesterolemia × 5 years

GERD × 15 years

FH

The patient is the oldest of seven children (four brothers and two sisters), all alive and well. He has two children, both in good health. Family history of terminal prostate cancer in his father (died at age 63). No other history of malignancy that he is aware of.

SH

The patient is employed by a large chemical company as a manager of the division that caters to aerospace applications. He has been with the company since graduating from college. He initially began work in the production areas and advanced to his current management position. He previously smoked 1–2 ppd for 25 years; he quit when he was diagnosed with GERD. He drinks one to two beers nightly. Diet is mostly unremarkable except that he states that he does not eat many vegetables, unless popcorn counts. He has been married for 34 years; his wife is with him today in the clinic.

ROS

The patient reports continuing fever, typically ranging from 100.2 to 101°F (37.9–38.3°C). He states his lower extremity swelling is much better since hospitalization and was pleased that he was able to wear his fancy dancing shoes for his daughter's wedding. He reports an unexplained weight loss of approximately 25 lb over the past 3 months. He denies headaches, changes in vision, or fainting episodes. He reports no lesions in his mouth, difficulty swallowing, or nosebleeds. He denies orthopnea and tachycardia. He has not had any back pain. He denies any changes in urination frequency, burning on urination, urinary frequency, dribbling, or blood in the urine. He has not noticed any abnormal bleeding or bruising. He has not received any prior transfusions.

Meds

Lisinopril 20 mg PO once daily

Furosemide 20 mg PO once daily

Simvastatin 20 mg PO at bedtime

Famotidine 20 mg PO twice daily

Temazepam 30 mg PO at bedtime as needed

Epoetin alfa 40,000 units subQ once weekly

All

Penicillin—rash

Physical Examination

Gen

Patient is a thin white man in no apparent distress.

VS

BP 145/100 mm Hg, P 95 bpm, RR 14, T 37.9°C; Wt 72 kg, Ht 5'9"

Skin

No rashes or moles noted

HEENT

PERRLA; TMs clear; no masses in the tonsils, palate, or floor of the mouth; no stomatitis.

Several missing teeth, but no gingival inflammation is noted.

Neck

Supple; no masses; no JVD

Chest

CTA bilaterally

CV

RRR; no MRG

Abd

Soft, nontender, appears distended. No hepatomegaly or splenomegaly noted. Bowel sounds normoactive.

Genit/Rect

Normal male genitalia

Ext

1+ pitting edema of le_ lower extremity

Neuro

Symmetric cranial nerve function. Symmetric facial muscle movement, and the tongue is midline. The palate is symmetric. Balance and coordination of the upper extremities are intact, with no evidence of tremor. There is symmetric coordination of rapidly alternating movements. Motor strength in the upper and lower extremities is normal and symmetric.

Lymph Node Survey

The lymph node survey is negative for any palpable peripheral nodes in the preauricular, postauricular, submandibular, cervical, supraclavicular, infraclavicular, or axillary areas.

Labs

Na 132 mEq/L	Hgb 10.3 g/dL	AST 29 IU/L	Phos 4.0 mg/dL
K 4.6 mEq/L	Hct 30%	ALT 27 IU/L	Uric acid 5.6 mg/dL
Cl 97 mEq/L	Plt $338 \times 10^3/\text{mm}^3$	Alk phos 75 IU/L	PT 12.2 seconds
CO ₂ 26 mEq/L	WBC $9.9 \times 10^3/\text{mm}^3$	LDH 623 IU/L	aPTT 21.7 seconds
BUN 20 mg/dL	Neutros 70%	T. bili 0.6 mg/dL	
SCr 0.9 mg/dL	Bands 2%	T. prot 6.3 g/dL	
Glu 112 mg/dL	Lymphs 18%	Alb 3.7 g/dL	
Ca 9.7 mg/dL	Monos 9%		
	Eos 1%		

Bone Marrow Biopsy

Biopsy and aspiration were negative for blasts or atypical lymphocytes.

Tumor Pathology

Diffuse large B-cell lymphoma, germinal center type; CD 10+, CD19 +, CD20+, CD23+, CD38+, CD5-. FISH indicates 75% of nuclei have MYC/IGH fusion and BCL2 rearrangement.

Initial Assessment

Diffuse large cell lymphoma. Further staging will include PET scan, HIV test, CT of the abdomen, and a baseline cardiac assessment in light of the patient's longstanding history of HTN.

Clinical Course

PET scanning revealed multiple foci of increased FDG uptake, consistent with findings of CT scan. No additional areas of involvement noted. MUGA scan reveals LVEF 65%.

Assessment

Double-hit diffuse large B-cell lymphoma, stage III; IPI score of 2.

QUESTIONS

1. What subjective and objective information is consistent with the diagnosis of non-Hodgkin lymphoma (NHL)?
2. Explain what system of staging was used and how his stage of disease was determined.
3. Create an individualized, patient-centered, team-based care plan to optimize medication therapy for this patient's NHL. Include specific drugs, dosage forms, doses, schedules, and durations of therapy.
4. What other interventions should be made to maintain control of the patient's other concurrent diseases?
5. How is the response to the treatment regimen for lymphoma assessed?

CLINICAL COURSE (PART 1)

The patient tolerated the first few cycles of chemotherapy well, with only some minimal nausea and vomiting. His antihypertensive medication was modified, increasing the lisinopril to 40 mg daily and achieving average systolic BPs in the 120s and average diastolic BPs in the 70s. His left lower extremity edema resolved, and furosemide was discontinued. His fasting lipid panel was checked and was found to be within his goals. One week after completing the fourth cycle of chemotherapy, he presented to the ED with fever (101.3°F [38.5°C]), cough, dyspnea, pain on inspiration, and fatigue. Laboratory evaluation showed an ANC of $0.352 \times 10^3/\text{mm}^3$. He was admitted to the hospital for evaluation and treatment of suspected neutropenic fever with pneumonia. Blood and sputum cultures were negative. The patient was treated with broad-spectrum antibiotics and became afebrile after 3 days. He was discharged from the hospital after the neutropenia resolved, completing a 14-day course of inpatient IV antibiotics. PET and CT scans performed while he was hospitalized to evaluate his lymphoma showed that he achieved a complete response.

FOLLOW-UP QUESTION

6. What measures should be taken to prevent neutropenic fever in subsequent courses of chemotherapy?

CLINICAL COURSE (PART 2)

The patient completed his planned course of chemotherapy without further event. Eighteen months later, he returns to the oncologist office after being diagnosed with relapsed lymphoma during a hospital admission for worsening dyspnea.

GENERAL QUESTIONS

7. What is the role of stem cell transplant in the treatment of aggressive NHL?
8. What therapeutic options are available for the treatment of relapsed diffuse large B-cell lymphomas?
9. If the patient had a history of hepatitis B, what diagnostic tests would be indicated? What antiviral therapy, if any, should be considered for patients with a history of hepatitis B who are receiving chemoimmunotherapy for treatment of diffuse large B-cell lymphoma?

